

Homework 3

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group 914.

① Find the sum for each of the following series.

$$a) \sum_{n \geq 2} \ln\left(1 - \frac{1}{n^2}\right) = \sum_{n \geq 2} x_n$$

$$x_n = \ln\left(1 - \frac{1}{n^2}\right) = \ln\left(\frac{n^2 - 1}{n^2}\right) = \ln\left(\frac{(n-1)(n+1)}{n^2}\right) =$$

$$= \ln(n-1) + \ln(n+1) - \ln(n^2) = \ln(n-1) + \ln(n+1) - 2 \ln n.$$

$$\sum_{n \geq 2} x_n = \sum_{n \geq 2} \ln(n-1) + \sum_{n \geq 2} \ln(n+1) - 2 \sum_{n \geq 2} \ln n$$

$$\text{Partial Sum: } S_N = \underbrace{\sum_{n \geq 2}^N \ln(n-1)}_{S_1} + \underbrace{\sum_{n \geq 2}^N \ln(n+1)}_{S_2} - 2 \cdot \underbrace{\sum_{n \geq 2}^N \ln(n)}_{S_3}$$

$$S_N = S_1 + S_2 + S_3.$$

$$S_1 = \ln 1 + \ln 2 + \ln 3 + \dots + \ln(N-1)$$

$$S_2 = \ln 3 + \ln 4 + \ln 5 + \dots + \ln(N+1)$$

$$S_3 = -2 \cdot (\ln 2 + \ln 3 + \ln 4 + \ln 5 + \dots + \ln(N))$$

$$\begin{aligned} \Rightarrow S_N &= \ln 1 + \ln 2 + \cancel{2 \ln 3} + \cancel{2 \ln 4} + \cancel{2 \ln 5} + \dots + \cancel{2 \ln(N-1)} + \ln N + \ln(N+1) \\ &\quad - \cancel{2 \ln 2} - \cancel{2 \ln 3} - \cancel{2 \ln 4} - \cancel{2 \ln 5} - \dots - \cancel{2 \ln(N-1)} - 2 \ln N \\ &= \ln 1 + \ln 2 + \ln N + \ln(N+1) - 2 \ln 2 - 2 \ln N \\ &= \ln 1 - \ln 2 - \ln N + \ln(N+1) \\ &= \ln 1 + \ln(N+1) - (\ln 2 + \ln N) \\ &= \ln \frac{N+1}{2N} \end{aligned}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \ln \frac{N+1}{2N} = \ln \frac{1}{2} = \ln 2^{-1} = -\ln 2.$$

$$b) \sum_{n \geq 1} \frac{n+1}{3^n} = \underbrace{\sum_{n \geq 1} \frac{n}{3^n}}_{S_1} + \underbrace{\sum_{n \geq 1} \frac{1}{3^n}}_{S_2}$$

$$S_1 = \sum_{n \geq 1} \frac{n}{3^n} = \frac{1}{3^1} + \frac{2}{3^2} + \frac{3}{3^3} + \dots + \frac{n}{3^n} + \dots$$

$$= \frac{1}{3} + \frac{1}{3} \left(\frac{2}{3} + \frac{3}{3^2} + \dots + \frac{n}{3^{n-1}} + \frac{n+1}{3^n} + \dots \right)$$

$$= \frac{1}{3} + \frac{1}{3} \cdot \left(\frac{2-1+1}{3^{2-1}} + \frac{3-1+1}{3^{3-1}} + \dots + \frac{n-1+1}{3^{n-1}} + \dots \right)$$

$$= \frac{1}{3} + \frac{1}{3} \cdot \left(\underbrace{\frac{1}{3} + \frac{1}{3} + \frac{2}{3^2} + \frac{1}{3^2} + \dots + \frac{n-1}{3^{n-1}} + \frac{1}{3^{n-1}} + \dots}_{S_1 + S_2} \right)$$

$$= \frac{1}{3} + \frac{1}{3} \cdot \left(\underbrace{\frac{1}{3} + \frac{2}{3^2} + \dots + \frac{n-1}{3^{n-1}}}_{S_1} + \underbrace{\frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}}}_{S_2} \right)$$

$$= \frac{1}{3} + \frac{1}{3} \cdot (S_1 + S_2)$$

$$S_2 = S_3 = \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}} = \frac{1}{3} \left(1 + \frac{1}{3} + \dots + \frac{1}{3^{n-1}} + \frac{1}{3^n} + \dots \right)$$

$$\stackrel{\text{geometrische Reihe}}{=} \frac{1}{3} \cdot \frac{1}{1 - \frac{1}{3}} = \frac{1}{3} \cdot \frac{1}{\frac{2}{3}} = \frac{1}{2}$$

$$\Rightarrow S_1 = \frac{1}{3} + \frac{1}{3} \cdot S_1 + \frac{1}{3} \cdot \frac{1}{2} \Rightarrow \frac{2S_1}{3} = \frac{1}{2} \Rightarrow S_1 = \frac{3}{4}$$

$$\Rightarrow \sum_{n \geq 1} \frac{n+1}{3^n} = S_1 + S_2 = \frac{3}{4} + \frac{1}{2} = \frac{5}{4}$$

$$c) \sum_{n \geq 1} \frac{n}{n^4 + n^2 + 1} = \sum_{n \geq 1} \frac{1}{(n^2+1)^2 - n^2} = \sum_{n \geq 1} \frac{1}{(n^2+n+1)(n^2-n+1)} =$$

$$= \frac{1}{2} \cdot \sum_{n \geq 1} \frac{(n^2+n+1) - (n^2-n+1)}{(n^2+n+1)(n^2-n+1)} = \frac{1}{2} \cdot \sum_{n \geq 1} \left(\frac{1}{n^2+n+1} - \frac{1}{n^2-n+1} \right)$$

$$= \frac{1}{2} \cdot \left(\begin{array}{l} \frac{1}{1} - \frac{1}{3} \\ + \frac{1}{3} - \frac{1}{7} \\ + \dots \\ + \frac{1}{(n-1)^2 - (n-1) + 1} - \frac{1}{(n-1)^2 + (n-1) + 1} \\ + \frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1} \\ + \dots \end{array} \right) = \frac{1}{2} \cdot \left(1 - \frac{1}{n^2+n+1} + \dots \right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{2} \left(1 - \frac{1}{n^2+n+1} \right) = \frac{1}{2} (1 - 0) = \frac{1}{2}$$

② Study the convergence of the following series:

a) $\sum_{n=1}^{\infty} \frac{x^n}{n^p}$, $x > 0$, $p \in \mathbb{N}$.

$$x_n = \frac{x^n}{n^p}$$

$$y_n = \frac{1}{n^p} \begin{cases} \text{divergent when } p \in \{0, 1\} \\ \text{convergent when } p > 1. \end{cases}$$

$$\lim_n \frac{x_n}{y_n} = \lim_n \frac{\frac{x^n}{n^p}}{\frac{1}{n^p}} = \lim_n \frac{x^n}{1} \stackrel{\geq 1}{> 0} \in (0, \infty), (\forall) x > 0, p \in \mathbb{N}$$

$$\lim_n x^n = \begin{cases} 0, & x \in (0, 1) \\ 1, & x = 1 \\ \infty, & x \in (1, \infty) \end{cases}$$

Case 1: $p \in \{0, 1\}$, $x \in (0, 1)$

$\Rightarrow y_n$ divergent

$\Rightarrow \lim_n \frac{x_n}{y_n} = 0$? $\rightarrow$ Comparing with x^n :

$$x_n = \frac{x^n}{n^p} \leq x^n; \quad x^n \text{ converges (ratio)} \Rightarrow x_n = \frac{x^n}{n^p} \text{ converges.}$$

Case 2: $p \in \{0, 1\}$, $x = 1$

$\Rightarrow y_n$ divergent

$$\Rightarrow \lim_n \frac{x_n}{y_n} = 1 \in (0, \infty)$$

$\Rightarrow x_n$ is also divergent.

Case 3: $p \in \{0, 1\}$, $x \in (1, \infty)$

$\Rightarrow y_n$ divergent

$$\Rightarrow \lim_n \frac{x_n}{y_n} = \infty$$

$\Rightarrow x_n$ is also divergent.

Case 4: $p > 1$, $x \in (0, 1)$

$\Rightarrow y_n$ convergent.

$$\Rightarrow \lim_n \frac{x_n}{y_n} = 0$$

$\Rightarrow x_n$ is also convergent.

Case 5: $p > 1, x = 1.$

$\Rightarrow y_n$ convergent

$\Rightarrow \lim_n \frac{x_n}{y_n} = 1 \in (0, \infty)$

$\Rightarrow x_n$ is also convergent.

Case 6: $p > 1, x \in (1, \infty)$

$\Rightarrow y_n$ convergent

? $\rightarrow$ Ratio test:

$\Rightarrow \lim_n \frac{x_n}{y_n} = \infty$

$$\frac{x_{n+1}}{x_n} = \frac{x^{n+1} \cdot x}{(n+1)^p} \cdot \frac{n^p}{x^n} = x \cdot \left(\frac{n}{n+1}\right)^p \rightarrow \infty \cdot 1 = \infty \Rightarrow \text{divergent.}$$

Conclusion: $\sum_{n \geq 1} \frac{x^n}{n^p} \rightarrow$ convergent, $(\forall) x \in (0, 1], (\forall) p \in \mathbb{N}.$
 $\searrow$
 divergent,

b) $\sum_{n \geq 2} \frac{1}{(\ln n)^{\ln n}} = \sum_{n \geq 2} \left(\frac{1}{\ln n}\right)^{\ln n} = \sum_{n \geq 2} x_n.$

~~$\ln n \rightarrow \ln 2 \rightarrow 0$~~

~~$\ln n < n^{-1}$~~

~~$\frac{1}{\ln n} > \frac{1}{n} \mid 1(\ln n) > 0$~~

~~$\frac{1}{(\ln n)^{\ln n}} > \frac{1}{n \ln n}$~~

$\sum_{n \geq 2} y_n = \sum_{n \geq 2} \frac{1}{n^2}$ diverges converges.

$\lim_n \frac{x_n}{y_n} = \lim_n \frac{\frac{1}{(\ln n)^{\ln n}}}{\frac{1}{n^2}} =$

$= \lim_n \frac{n^2}{(\ln n)^{\ln n}} =$

$= \lim_n \frac{n^2}{\left(e^{\ln(\ln n)}\right)^{\ln n}} =$

$= \lim_n \frac{e^{2 \ln n}}{e^{\ln(n) \cdot \ln(\ln n))}} = \lim_n \left(e^{2 \ln n - \ln(n) \cdot \ln(\ln n)} \right) =$

$= e^{\lim_n \ln n (2 - \ln(\ln n))} \stackrel{(*)}{=} e^{\infty \cdot (-\infty)} = e^{-\infty} = \frac{1}{e^{\infty}} = 0. \Rightarrow$

$\Rightarrow x_n$ also converges.

$\Rightarrow \sum_{n \geq 2} x_n$ convergent.

$\lim_n (2 - \ln(\ln n)) =$

$= 2 - \infty = -\infty \quad (*)$

$p \in [0, 1], x \in (0, 1]$

$p > 1$

$(\forall) p \in \mathbb{N}$

$p \in [0, 1] \quad x \in [1, \infty)$

$p > 1$

$$c) \sum_{n \geq 1} (\sqrt[n]{n} - 1) = \sum_{n \geq 1} (n^{\frac{1}{n}} - 1) = \sum_{n \geq 1} x_n.$$

$$\sum_{n \geq 1} y_n = \sum_{n \geq 1} \frac{1}{n} \text{ diverges.}$$

$$\begin{aligned} \lim_n \frac{x_n}{y_n} &= \lim_n \frac{n^{\frac{1}{n}} - 1}{\frac{1}{n}} = \lim_n n \cdot (n^{\frac{1}{n}} - 1) = \\ &= \lim_n n \cdot (e^{\frac{\ln n}{n}} - 1) = \lim_n \frac{e^{\frac{\ln n}{n}} - 1}{\frac{\ln n}{n}} \cdot \ln n = \lim_n e^{\frac{\ln n}{n}} \cdot \lim_n \ln n \\ &= \infty \end{aligned}$$

$$\text{Let } x = \frac{\ln n}{n}.$$

$$e^x = \exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots$$

$$\Rightarrow \lim_n \frac{x_n}{y_n} = \infty \Rightarrow \sum_{n \geq 1} x_n \text{ is also divergent.}$$

$$d) \sum_{m \geq 1} \frac{1 \cdot 3 \cdot \dots \cdot (2m-1)}{2 \cdot 4 \cdot \dots \cdot 2m} \cdot \frac{1}{2m+1} = \sum_{m \geq 1} x_m$$

$$\frac{2(m+1)-1}{2m+2-1} = \frac{2m+1}{2m+1}$$

$$\text{Ratio test: } \lim_n \frac{x_{m+1}}{x_m} = \lim_n \frac{1 \cdot 3 \cdot \dots \cdot (2m-1) \cdot (2m+1)}{2 \cdot 4 \cdot \dots \cdot (2m+2) \cdot (2m+3)} \cdot \frac{2 \cdot 4 \cdot \dots \cdot 2m \cdot (2m+1)}{1 \cdot 3 \cdot \dots \cdot (2m-1)},$$

$$\lim_n \frac{(2m+1)^2}{(2m+2)(2m+3)} = 1 \Rightarrow \text{Raabe - Dehamel:}$$

$$\lim_n m \cdot \left(\frac{x_m}{x_{m+1}} - 1 \right) =$$

$$= \lim_n m \cdot \left(\frac{1 \cdot 3 \cdot \dots \cdot (2m-1)}{2 \cdot 4 \cdot \dots \cdot 2m \cdot (2m+1)} \cdot \frac{2 \cdot 4 \cdot \dots \cdot 2m \cdot (2m+2) \cdot (2m+3)}{1 \cdot 3 \cdot \dots \cdot (2m-1) \cdot (2m+1)} - 1 \right)$$

$$= \lim_n m \cdot \left(\frac{(2m+2)(2m+3)}{(2m+1)^2} - 1 \right)$$

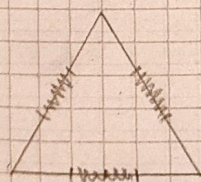
$$= \lim_n m \cdot \left(\frac{4m^2 + 10m + 6 - (2m+1)^2}{(2m+1)^2} \right)$$

$$= \lim_n m \cdot \left(\frac{4m^2 + 10m + 6 - 4m^2 - 4m - 1}{4m^2 + 4m + 1} \right)$$

$$= \lim_n m \cdot \left(\frac{6m + 5}{4m^2 + 4m + 1} \right) = \frac{6}{4} = \frac{3}{2} > 1 \Rightarrow \text{convergent.}$$

$$\Rightarrow \sum_{m \geq 1} x_m \text{ convergent.}$$

3)



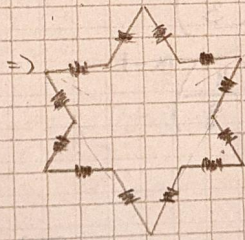
$$l = 1$$

3 sides

$$P_0 = 3 \cdot 1 = 3$$



3 \cdot (3+1) sides



$$l = \frac{1}{3}$$

3 \cdot (3+1) sides

$$P_1 = 3 \cdot (3+1) \cdot \frac{1}{3}$$



(3+1) \cdot 3 \cdot (3+1) sides

\Rightarrow

$$l = \frac{1}{3}$$

(1+3) \cdot (3 \cdot (3+1)) sides

$$P_2 = (3+1) \cdot (3 \cdot (3+1)) \cdot \frac{1}{3}$$



(3+1) \cdot (3+1) \cdot (3 \cdot (3+1)) sides

$$\Rightarrow \text{Number of sides: } 3 \cdot 4^m = N(m)$$

$$\Rightarrow \text{Side length: } \frac{1}{3^m} = S(m)$$

$$\Rightarrow \text{Perimeter: side length} \times \text{number of sides} = \frac{1}{3^m} \cdot 3 \cdot 4^m = P(m)$$

$$\Rightarrow \text{Area: } \frac{\text{side length} \times \sqrt{3}}{4} = \frac{1}{4} \cdot \frac{\sqrt{3}}{3^m}$$

$$\lim_n P_n = \lim_n 3 \cdot \left(\frac{4}{3}\right)^n = \infty, \text{ diverges.}$$

↓
>1

~~$$\lim_n A_n = \lim_n \frac{1}{3^n} \cdot \sqrt{3}$$~~

~~$$A_n = A_{n-1} + n \cdot \text{sides}$$~~

~~$$A_n = A_{n-1} + \text{Side-length}(A_n)$$~~

$$A_n = A_{n-1} + N_{n-1} \cdot \frac{S_n^2 \cdot \sqrt{3}}{4}$$

$$N_n = 3 \cdot 4^n$$

$$S_n = \frac{1}{3^n}$$

$$A_0 = \frac{\sqrt{3}}{4}$$

$$\begin{aligned} \Rightarrow A_n &= A_{n-1} + 3 \cdot 4^{n-1} \cdot \frac{\left(\frac{1}{3^n}\right)^2 \cdot \sqrt{3}}{4} \\ &= A_{n-1} + 3 \cdot 4^{n-1} \cdot \frac{1}{32^n} \cdot A_0 \\ &= A_{n-1} + A_0 \cdot 3 \cdot 4^{n-1} \cdot \frac{1}{3^{n-1} \cdot 3^{n+1}} \\ &= A_{n-1} + A_0 \cdot \frac{3}{3^{n+1}} \cdot \left(\frac{4}{3}\right)^{n-1} \\ &= A_{n-1} + A_0 \cdot \frac{1}{3^n} \cdot \left(\frac{4}{3}\right)^{n-1} \end{aligned}$$

$$\begin{aligned} A_n &= A_0 + \sum_{k=1}^n A_0 \cdot \frac{1}{3} \cdot \left(\frac{4}{3}\right)^{k-1} \\ &= A_0 \cdot \left(1 + \frac{1}{3} \cdot \sum_{k=0}^{n-1} \left(\frac{4}{3}\right)^k\right) \\ &= A_0 \cdot \left(1 + \frac{1}{3} \cdot \frac{1 - \left(\frac{4}{3}\right)^n}{1 - \frac{4}{3}}\right) \\ &= A_0 \cdot \left(1 + \frac{3}{5} \left(1 - \left(\frac{4}{3}\right)^n\right)\right) \end{aligned}$$

↓
>40

$$\Rightarrow \lim_n A_n = \lim_n A_0 \cdot \left(1 + \frac{3}{5} (1 - 0)\right) =$$

$$= \lim_n A_0 \cdot \frac{8}{5} = \frac{\sqrt{3}}{4} \cdot \frac{8}{5} = \frac{\sqrt{3}}{4} \cdot \frac{8}{5} = \frac{2\sqrt{3}}{5}, \text{ converges}$$

$$\sum_{k=0}^{n-1} \left(\frac{4}{3}\right)^k = \frac{1 - \left(\frac{4}{3}\right)^n}{1 - \frac{4}{3}}$$